

Internal verification of assessment decisions – BTEC Higher Nationals (RQF)

INTERNAL VERIFICATION – ASSESSMENT DECISIONS (Single Student)					
Programme Title:					
Unit Number and Title:					
Assessor Name:				Internal Verifier Name:	
Assignment title:					
Name of student	Submission Type (First, Resubmission, Retake)	Grade the Assessor has awarded.	Assessment Decision Accurate (Y/N) (Resubmission and retake must be capped at a Pass)	List the learning outcomes and grading criteria where inaccurate decisions have been made	State why the assessment decision is inaccurate. <i>*If an inaccurate decision is recorded the Internal Verifier must recommend actions detailing the issues to be addressed. The Assessor and the Internal Verifier must then confirm that the action has been undertaken before assessment decisions are issued to the student.</i>
INTERNAL VERIFIER CHECKLIST					Y/N
Has the student and the Assessor confirmed the authenticity of the evidence?					
Is there evidence of collusion or plagiarism? The work submitted for assessment has been carried out without assistance other than that which is acceptable according to the rules of the specification. The evidence submitted for this assignment is student's own. The student has clearly referenced any sources and any artificial intelligence (AI) tools used in the work.					
Is the assessor feedback to the student appropriate and constructive to each student? Provide comments below. • Points out strengths and areas for improvement. • Linked to relevant learning outcomes and assessment criteria. • Clear as to why the student did not achieve higher grades. • Identify opportunities for improved performance in future assignments.					

GENERAL COMMENTS

Any actions required must be reviewed across the whole cohort.			
Action Required		Target Date for Completion	Date Action Completed
I confirm that the assessment decisions are accurate, there is no evidence of assessment malpractice and any action points have been addressed and completed in respect of the whole cohort. I declare the work submitted for assessment has been carried out without assistance other than that which is acceptable according to the rules of the specification. The evidence submitted for this assignment is student's own. The student has clearly referenced any sources and any artificial intelligence (AI) tools used in the work.			
Internal Verifier signature		Date	
Assessor signature		Date	

Higher Nationals - Summative Assignment Feedback Form

Student Name/ID			
Unit Title	Discrete Maths		
Assignment Number	Assignment 01	Assessor	
Submission Date		Date Received 1st submission	
Re-submission Date		Date Received 2nd submission	
Assessor Feedback: *Please note that constructive and useful feedback should allow students to understand: a) Strengths of performance b) Limitations of performance c) Any improvements needed in future assessments Feedback should be against the learning outcomes and assessment criteria to help students understand how these inform the process of judging the overall grade. Feedback should give full guidance to the students on how they have met the learning outcomes and assessment criteria.			
<i>I certify that to the best of my knowledge the evidence submitted for this assignment is the student's own. The student has clearly referenced any sources and any artificial intelligence (AI) tools used in the work. I have not solely used AI to grade the student's work.</i>			
Grade:	Assessor Signature:		Date:
Resubmission Feedback: *Please note resubmission feedback is focussed only on the resubmitted work			
Grade:	Assessor Signature:		Date:
Internal Verifier's Comments:			
Signature & Date:			

* Please note that grade decisions are provisional. They are only confirmed once internal and external moderation has taken place and grades decisions have been agreed at the assessment board.

STUDENT ASSESSMENT SUBMISSION AND DECLARATION

When submitting evidence for assessment, each student must sign a declaration confirming that the work is their own.

Student name: MN. Sihanas		Assessor name: Eng. A.L. Jubailah Begum	
Issue date: 03.03.2025	Submission date: 13.04.2025		Submitted on: 10.04.2025
Programme: Pearson BTEC Higher Nationals in Computing			
Unit: Discrete Maths			
Assignment number and title: 1. Applications for sets, graph theory, Booleans & abstract algebra			

Plagiarism

Plagiarism is a particular form of cheating. Plagiarism must be avoided at all costs and students who break the rules, however innocently, may be penalised. It is your responsibility to ensure that you understand correct referencing practices. As a university level student, you are expected to use appropriate references throughout and keep carefully detailed notes of all your sources of materials for material you have used in your work, including any material downloaded from the Internet. Please consult the relevant unit lecturer or your course tutor if you need any further advice.

Student Declaration

Student declaration I certify that the assignment submission is entirely my own work and I fully understand the consequences of plagiarism. I declare that the work submitted for assessment has been carried out without assistance other than that which is acceptable according to the rules of the specification. I certify I have clearly referenced any sources and any artificial intelligence (AI) tools used in the work. I understand that making a false declaration is a form of malpractice.	
Student signature: 1030371@bcas.ac	Date: 2025.04.13

General Guidelines

1. A Cover page or title page – You should always attach a title page to your assignment. Use previous page as your cover sheet and make sure all the details are accurately filled.
2. Attach this brief as the first section of your assignment.
3. All the assignments should be prepared using a word processing software.
4. All the assignments should be printed on A4 sized papers. Use single side printing.
5. Allow 1” for top, bottom, right margins and 1.25” for the left margin of each page.

Word Processing Rules

1. The font size should be 11 points, and should be in the style of Arial.
2. Use 1.5 line spacing. Left justify all paragraphs.
3. Ensure that all the headings are consistent in terms of the font size and font style.
4. Use footer function in the word processor to insert Your Name, Subject, Assignment No, and Page Number on each page. This is useful if individual sheets become detached for any reason.
5. Use word processing application spell check and grammar check function to help editing your assignment.

Important Points

1. It is strictly prohibited to use text boxes to add texts in the assignments, except for the compulsory information. eg: Figures, tables of comparison etc. Adding text boxes in the body except for the before mentioned compulsory information will result in rejection of your work.
2. Avoid using page borders in your assignment body.
3. Carefully check the hand in date and the instructions given in the assignment. Late submissions will not be accepted.
4. Ensure that you give yourself enough time to complete the assignment by the due date.
5. Excuses of any nature will not be accepted for failure to hand in the work on time.
6. You must take responsibility for managing your own time effectively.
7. If you are unable to hand in your assignment on time and have valid reasons such as illness, you may apply (in writing) for an extension.
8. Failure to achieve at least PASS criteria will result in a REFERRAL grade.
9. Non-submission of work without valid reasons will lead to an automatic REFERRAL. You will then be asked to complete an alternative assignment.
10. If you use other people's work or ideas in your assignment, reference them properly using HARVARD referencing system to avoid plagiarism. You have to provide both in-text citation and a reference list.
11. If you are proven to be guilty of plagiarism or any academic misconduct, your grade could be reduced to A REFERRAL or at worst you could be expelled from the course.

Unit 18: Discrete Maths

Assignment Brief

Programme Title	Pearson BTEC Higher Nationals in Computing
Student Name/ID Number	MN. Sihanass/1030371
Unit Number and Title	Unit 18: Discrete Maths
Academic Year	2025
Unit Tutor	Eng. A.L. Jubailah Begum
Assignment Title	Applications for sets, graph theory, Booleans & abstract algebra
Issue Date	03.03.2025
Submission Date	13.04.2025
Submission Format	
The submission should be in the form of an individual report written in a concise, formal business style using single spacing and font size 12. You are required to make use of headings, paragraphs and subsections as appropriate, and all work must be supported with research and referenced using Harvard referencing system. Please also provide an end list of references using the Harvard referencing system. The recommended word count is 5,000–5,500 words for the report excluding annexures, although you will not be penalised for exceeding the total word limit.	
Unit Learning Outcomes	
LO1. Examine set theory and functions applicable to software engineering. LO2. Analyze mathematical structures of objects using graph theory. LO3 Investigate solutions to problem situations using the application of Boolean algebra. LO4. Explore applicable concepts within abstract algebra.	
Transferable skills and competencies developed	
Vocational scenario	
You are assigned as a trainee in a software development center for a project and they want to check your knowledge in mathematics. The following questions are assigned to you; so that your supervisor can decide to which team you are suitable for.	
Assignment activity and guidance	

Q1)

i) Let A and B be two non-empty finite sets. If cardinalities of the sets A, B, and $A \cap B$ are 160, 50 and 25 respectively.

Find the cardinality of the set AUB.

If $n(A - B) = 30$, $n(A \cup B) = 70$ and $n(A \cap B) = 10$, then find $n(B)$.

If A, B and C are disjoint sets $n(A) = 200$, $n(B) = 140$ and $n(C) = 55$, find $n(A \cup B \cup C)$.

Note: You can use either Venn diagrams or set formulas / properties for the answer.

ii) Let A and B be two non-empty finite sets. If cardinalities of the sets A, B, and $A \cap B$ are 360, 150 and 75 respectively.

Find the cardinality of the set AUB.

If $n(A - B) = 50$, $n(A \cup B) = 170$ and $n(A \cap B) = 20$, then find $n(B)$.

If A, B and C are disjoint sets $n(A) = 800$, $n(B) = 340$ and $n(C) = 90$, find $n(A \cup B \cup C)$.

Note: You can use either Venn diagrams or set formulas / properties for the answer.

Q2) Find the inverse of the following functions.

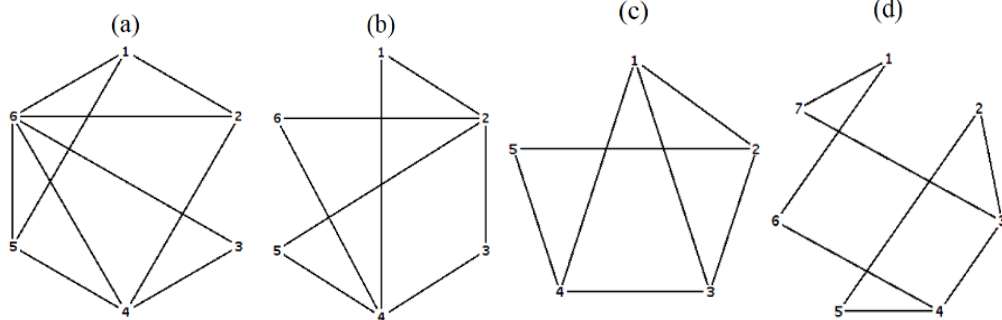
i) $f(x) = x + 60$

ii) $f(x) = 10 / (x+60)$

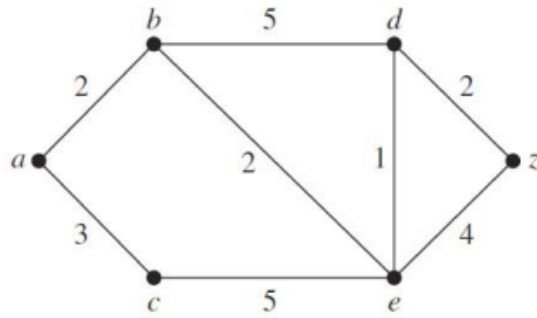
iii) $f(x) = (120-x) / 80$

iv) $f(x) = (x-5) / (x+5)$

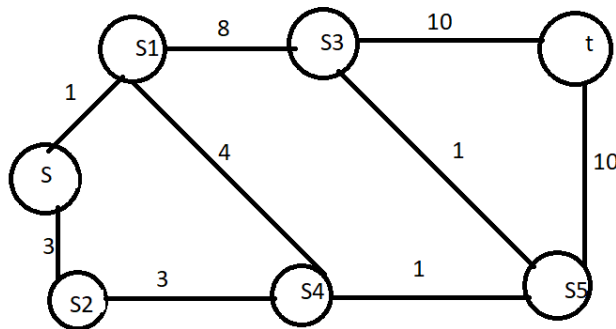
v) $f(x) = x^3 - 64$

Q3) Model contextualised problems using trees, both quantitatively and qualitatively.**Q4) Access whether an Eulerian and Hamiltonian circuit exists in the below given undirected graphs.****Q5)**

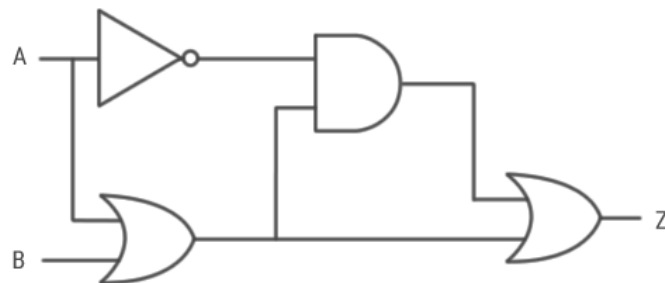
i) Find the shortest path and shortest distance from a to z of the following graph.



ii) Find the shortest path and shortest distance from s to t of the following graph.



Q6) Find the final output equation for the below logic circuit and prepare the truth table for that final output equation.



Q7)

Simplify below Boolean expressions.

- i) $AB + A(B+C) + B(B+C)$
- ii) $AB' + A(B+C)' + B(B+C)'$
- iii) $(AB'(C+BD) + A'B')C$
- iv) $A'BC + AB'C' + A'B'C' + AB'C + ABC$

Q8)

Create a truth table for the following given scenario and find the final output equation for that truth table using **SOP** method. Then, draw a logic gate based digital circuit for that final output equation.

There is an electronic door system. That system has two sensors called A and B. If at least one of the following conditions is satisfied, then the door is opened.

- Sensor A is activated.
- Both sensor A and Sensor B are activated.

Construct the truth table for above scenario.

The active state of the sensors is represented using 1.

The inactive state of the sensors is represented using 0.

Door open state is represented using 1.

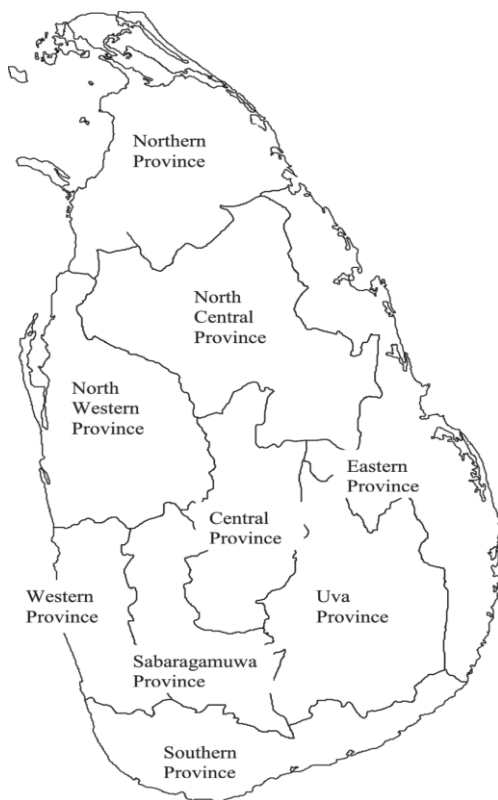
Door close state is represented using 0.

Q9)

Discuss a real-world binary problem in the application of Boolean algebra. Provide relevant diagrams.

Q10)

Colorize following provincial map of Sri Lanka using five-colour theorem.



Q11)

When considering the **set of all the natural numbers ($\mathbb{N}$)**, show whether the mathematical operations of **addition, subtraction, multiplication and division** are:

- (a) Associative binary operation
- (b) Commutative binary operation

Q12)

- (a) Determine the order of a group and the order of a subgroup in the group $G = \mathbb{Z}_6$ (the integers modulo 6 under addition).
- (b) Validate that the **set** of non-zero real numbers with the **operation of multiplication** is a commutative **group**.

Q13)

You are required to **prepare a 15-minutes presentation** aimed towards new students engaged under a technology company's graduate scheme.

Your presentation will include evidence of:

An explanation that adequately explains why **group theory** is taught to computing students.

Recommended Resources

Please note that the resources listed are examples for you to use as a starting point in your research – the list is not definitive.

Learning Outcomes and Assessment Criteria

Pass	Merit	Distinction
LO1 Examine set theory and functions applicable to software engineering		D1 Formulate corresponding proof principles to prove properties about defined sets.
P1 Perform algebraic set operations in a formulated mathematical problem. P2 Determine the cardinality of a given bag (multiset).	M1 Determine the inverse of a function using appropriate mathematical techniques.	
LO2 Analyse mathematical structures of objects using graph theory		D2 Construct a proof of the Five Colour Theorem.
P3 Model contextualized problems using trees, both quantitatively and qualitatively. P4 Use Dijkstra's algorithm to find a shortest path spanning tree in a graph.	M2 Assess whether an Eulerian and Hamiltonian circuit exists in an undirected graph.	
LO3 Investigate solutions to problem situations using the application of Boolean algebra		

P5 Diagram a binary problem in the application of Boolean Algebra. P6 Produce a truth table and its corresponding Boolean equation from an applicable scenario.	M3 Simplify a Boolean equation using algebraic methods.	D3 Design a complex system using logic gates.
LO4 Explore applicable concepts within abstract algebra		D4 Prepare a presentation that explains an application of group theory relevant to your course of study.
P7 Describe the distinguishing characteristics of different binary operations that are performed on the same set. P8 Determine the order of a group and the order of a subgroup in given examples.	M4 Validate whether a given set with a binary operation is indeed a group.	

Acknowledgement

I would like to express my sincere gratitude to my tutor, Eng. A.L. Jubailah Begum, for her invaluable guidance, motivation, and support throughout the completion of this assignment. Her detailed feedback and encouragement helped me to approach each problem with a deeper understanding and structured methodology. I also extend my thanks to BCAS Campus for providing an academic environment that fosters learning and innovation. Lastly, I am grateful to my peers and family for their constant encouragement during the course of this project.

Abstract

This report explores the essential mathematical concepts of set theory, graph theory, Boolean algebra, and abstract algebra, all of which are foundational to computing and software engineering. Through a series of real-world and theoretical problems, the assignment demonstrates the application of set operations, inverse functions, tree modelling, shortest path algorithms, logic circuits, Boolean simplifications, and group theory. The objective is to showcase how discrete mathematics provides critical tools for logical reasoning, system modelling, and problem-solving in the computing domain. Each question is tackled using precise mathematical techniques, supported by diagrams and step-by-step analysis, highlighting the relevance and practicality of discrete math in technical environments.

Table of content

Introduction	1
Q1 Finding the cardinalities	2
i)	2
Finding $nA \cup B$	2
Finding nB	2
Finding $nA \cup B \cup C$ when A, B, C are disjoint	3
ii)	4
Finding $n(A \cup B)$	4
Finding $n(B)$	4
Finding $n(A \cup B \cup C)$ when A, B, C are disjoint	5
Q2: Finding the inverse of the following functions.....	6
i) $f(x) = x + 60$	6
ii) $f(x) = 10/(x + 60)$	6
iii) $f(x) = (120 - x) / 80$	7
iv) $f(x) = (x - 5) / (x + 5)$	8
v) $f(x) = x^3 - 64$	8
Q3: Modelling contextualised problems using trees, both quantitatively and qualitatively.	9
3.1 Quantitative Example: Hierarchical Organization	9
3.1 Qualitative Example: Binary Search Tree (BST)	11
Q4: Access whether an Eulerian and Hamiltonian circuit exists in undirected graphs....	13
(a)	13
(b)	14
(c)	14
(d)	15
Q5: Finding the shortest path using Dijkstra's algorithm	16
i) Finding the shortest path and shortest distance from a to z of the following graph.	16
ii) Finding the shortest path and shortest distance from s to t of the following graph.	17
Q6: Finding the output equation for the logic circuit and prepare the truth table for the output equation.	18
Truth table for the output equation.....	18
Q7: Simplifying Boolean expressions.	19
i) $AB + A(B + C) + B(B + C)$	19
ii) $AB' + A(B + C)' + B(B + C)'$	19
iii) $(AB'(C + BD) + A'B')$	20
iv) $A'BC + AB'C' + A'B'C' + AB'C + ABC$	20
Q8: Truth Table Analysis and SOP Derivation for an Electronic Door System.	21
8.1 Truth Table:	21
8.2 Finding the final equation using Sum of Products (SOP)	21
8.3 logic gate based digital circuit for the final output equation.	21
Q9: real-world binary problem in the application of Boolean algebra	22

9.1 Example Scenario	22
9.2 Truth Table	22
9.3 Boolean Expression	23
9.4 Logic Circuit Diagram	23
9.5 Real-World Implementation.....	23
Q10: Colorize following provincial map of Sri Lanka using five-colour theorem.	24
Q11: analyze the properties of addition, subtraction, multiplication, and division concerning their associative and commutative nature.	26
(a) Associative Binary Operation	26
(b) Commutative Binary Operation	27
Q12:	28
(a) Determine the order of the group and the orders of its subgroups for $G = \mathbb{Z}_6$ (the integers modulo 6 under addition).	28
1. Order of the Group $\mathbb{Z}_6$:	28
2. Orders of Subgroups in $\mathbb{Z}_6$:	28
(b) Validate that the set of non-zero real numbers with the operation of multiplication is a commutative group.	29
Conclusion	30
References.....	31

List of tables

Table 1: Truth table of Logical circuit (Bratty, 2024)	18
Table 2: Truth table Electronic Door System (GeeksforGeeks, 2023).....	21
Table 3: Karnaugh Map (SOP) (GeeksforGeeks, 2015)	21
Table 4: Truth table for the Air Conditioner Control Scenario (Storr, 2013).....	22
Table 5: summary of Properties of Binary Arithmetic Operations (BrainRouter, 2020)...	27

List of figures

Figure 1: Quantitative Example (Hierarchical Organization)	10
Figure 2: Qualitative Example: Binary Search Tree (BST)	11
Figure 3: (a)Hamiltonian circuit path.....	13
Figure 4: (b)Eulerian circuit path	14
Figure 5: (c)Hamiltonian circuit path	14
Figure 6: (d)Hamiltonian circuit path.....	15
Figure 7: finding the shortest path from a to z.....	16
Figure 8: finding the shortest path from s to t	17
Figure 9: finding the logical circuit equation output of (z)	18
Figure 10: Logic circuit for Air Conditioner Control Scenario.....	23
Figure 11: colored provincial map of Sri Lanka using five-colour theorem.....	25

Introduction

Discrete Mathematics serves as a cornerstone for the field of computing, equipping students and professionals with tools to tackle logical and structural challenges in software design, algorithms, and systems analysis. This assignment delves into key areas including set theory, functions, graph theory, Boolean algebra, and abstract algebra, presenting a wide range of mathematical problems relevant to computing contexts.

The report begins with basic set operations and cardinalities, leading into inverse functions and graph structures. It proceeds to explore logic circuits, truth tables, and the practical use of Boolean expressions in systems such as security and automation. Finally, the assignment investigates the role of abstract algebra, particularly group theory, in underpinning logical operations in computing systems. Throughout the report, the emphasis is placed on analytical reasoning and the real-world applicability of mathematical concepts to computing scenarios.

Q1 Finding the cardinalities

i)

Finding $n(A \cup B)$

formula:

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

Given:

- $n(A) = 160$
- $n(B) = 50$
- $n(A \cap B) = 25$

Substituting the values:

$$n(A \cup B) = 160 + 50 - 25 = 185$$

Answer (SplashLearn, 2023):

$$n(A \cup B) = 185$$

Finding $n(B)$

Given:

- $n(A - B) = 30$
- $n(A \cup B) = 70$
- $n(A \cap B) = 10$

From set properties, we know:

$$n(A - B) = n(A) - n(A \cap B)$$

Substituting the given values:

$$30 = n(A) - 10 \quad n(A) = 40$$

$$n(A) = 40 \quad n(A) = 40$$

Now, using the union formula:

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$70 = 40 + n(B) - 10$$

$$n(B) = 40$$

Answer (SplashLearn, 2023):

$$n(B) = 40$$

Finding $n(A \cup B \cup C)$ when A, B, C are disjoint

For disjoint sets:

$$n(A \cup B \cup C) = n(A) + n(B) + n(C)$$

Given:

- $n(A) = 200$
- $n(B) = 140$
- $n(C) = 55$

Calculating:

$$n(A \cup B \cup C) = 200 + 140 + 55 = 395$$

Answer (SplashLearn, 2023):

$$n(A \cup B \cup C) = 395$$

ii)

Finding $n(A \cup B)$

Using the formula:

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

Given:

- $n(A) = 360$
- $n(B) = 150$
- $n(A \cap B) = 75$

Substituting the values:

$$n(A \cup B) = 360 + 150 - 75 = 435$$

Answer (SplashLearn, 2023):

$$n(A \cup B) = 435$$

Finding $n(B)$

Given:

- $n(A - B) = 50$
- $n(A \cup B) = 170$
- $n(A \cap B) = 20$

From set properties, we use:

$$n(A - B) = n(A) - n(A \cap B)$$

Substituting the given values:

$$50 = n(A) - 20$$

$$n(A) = 70$$

Now we use the union formula:

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$170 = 70 + n(B) - 20$$

$$n(B) = 120$$

Answer (SplashLearn, 2023):

$$n(B) = 120$$

Finding $n(A \cup B \cup C)$ when A, B, C are disjoint

For disjoint sets:

$$n(A \cup B \cup C) = n(A) + n(B) + n(C)$$

Given:

$$n(A) = 800$$

$$n(B) = 340$$

$$n(C) = 90$$

Calculating:

$$n(A \cup B \cup C) = 800 + 340 + 90 = 1230$$

Answer (SplashLearn, 2023):

$$n(A \cup B \cup C) = 1230$$

Q2: Finding the inverse of the following functions.

i) $f(x) = x + 60$

To find the inverse function $f^{-1}(x)$:

1. Replace $f(x)$ with y :

$$y = x + 60$$

2. Solve for x in terms of y :

$$x = y - 60$$

3. Swap x and y to get the inverse function:

$$f^{-1}(x) = x - 60$$

Answer (Dawkins, 2022):

$$f^{-1}(x) = x - 60$$

ii) $f(x) = 10/(x + 60)$

To find the inverse function $f^{-1}(x)$:

1. Replace $f(x)$ with y :

$$y = \frac{10}{x + 60}$$

2. Solve for x in terms of y :

$$y(x + 60) = 10$$

$$x + 60 = \frac{10}{y}$$

$$x = \frac{10}{y} - 60$$

3. Swap x and y to get the inverse function:

$$f^{-1}(x) = \frac{10}{x} - 60$$

Answer (Dawkins, 2022):

$$f^{-1}(x) = \frac{10}{x} - 60$$

iii) $f(x) = (120 - x) / 80$

To find the inverse function $f^{-1}(x)$:

1. Replace $f(x)$ with y :

$$y = \frac{120 - x}{80}$$

2. Solve for x in terms of y :

$$\begin{aligned} y \times 80 &= 120 - x \\ x &= 120 - 80y \end{aligned}$$

3. Swap x and y to get the inverse function:

$$f^{-1}(x) = 120 - 80x$$

Answer (Dawkins, 2022):

$$f^{-1}(x) = 120 - 80x$$

iv) $f(x) = (x - 5) / (x + 5)$

To find the inverse function $f^{-1}(x)$:

1. Replace $f(x)$ with y :

$$y = \frac{x - 5}{x + 5}$$

2. Solve for x in terms of y :

$$y(x + 5) = x - 5$$

$$yx + 5y = x - 5$$

$$yx - x = -5 - 5y$$

$$x(y - 1) = -5 - 5y$$

$$x = \frac{(-5 - 5y)}{y - 1}$$

3. Swap x and y to get the inverse function:

$$f^{-1}(x) = \frac{(-5 - 5x)}{x - 1}$$

Answer (Dawkins, 2022):

$$f^{-1}(x) = \frac{(-5 - 5x)}{x - 1}$$

v) $f(x) = x^3 - 64$

To find the inverse function $f^{-1}(x)$:

1. Replace $f(x)$ with y :

$$y = x^3 - 64$$

2. Solve for x in terms of y :

$$x^3 = y + 64$$

$$x = \sqrt[3]{y + 64}$$

3. Swap x and y to get the inverse function:

$$f^{-1}(x) = \sqrt[3]{x + 64}$$

Answer (Dawkins, 2022):

$$f^{-1}(x) = \sqrt[3]{x + 64}$$

Q3: Modelling contextualised problems using trees, both quantitatively and qualitatively.

A **binary tree** is a hierarchical data structure in which each node has at most two children, commonly referred to as the left and right child. This structure is fundamental in computer science and discrete mathematics due to its efficiency in organizing and managing data (Weisstein, 2025).

3.1 Quantitative Example: Hierarchical Organization

Consider a company's organizational chart, which can be represented as a binary tree:

- **Level 0 (Root):** The CEO is at the top, serving as the root node.
- **Level 1:** The CEO has two direct reports: the Chief Operating Officer (COO) on the left and the Chief Financial Officer (CFO) on the right.
- **Level 2:** Each of these officers oversees two departments. For instance, the COO supervises the Heads of Operations and Human Resources, while the CFO oversees the Heads of Accounting and Treasury.

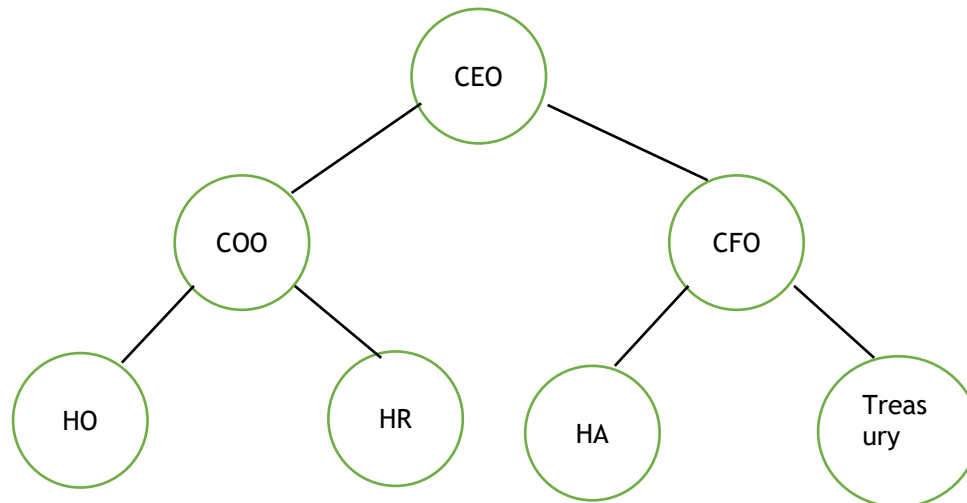


Figure 1: Quantitative Example (Hierarchical Organization)

Types of Binary Trees:

1. **Full Binary Tree:** Every node has either 0 or 2 children.
2. **Complete Binary Tree:** All levels are fully filled except possibly the last, which is filled from left to right.
3. **Perfect Binary Tree:** All internal nodes have two children, and all leaves are at the same level.
4. **Balanced Binary Tree:** The height of the left and right subtrees of any node differ by at most one (Weisstein, 2025).

This binary tree structure effectively models the hierarchical relationships within the company, illustrating clear reporting lines and organizational flow (Weisstein, 2025).

3.1 Qualitative Example: Binary Search Tree (BST)

A Binary Search Tree is a specialized form of a binary tree that facilitates efficient data retrieval. In a BST, each node adheres to the following properties:

- The left subtree contains nodes with values less than the parent node's value.
- The right subtree contains nodes with values greater than the parent node's value.

This arrangement allows for operations like insertion, deletion, and lookup to be performed in logarithmic time complexity, $O(\log n)$, under optimal conditions (Geeksforgeeks, 2015).

Constructing a BST:

Given a set of numbers: [15, 10, 20, 8, 12, 17, 25], we can construct the following BST:

1. **Insert 15** as the root node.
2. **Insert 10**: Since $10 < 15$, it becomes the left child of 15.
3. **Insert 20**: Since $20 > 15$, it becomes the right child of 15.
4. **Insert 8**: Since $8 < 15$ and $8 < 10$, it becomes the left child of 10.
5. **Insert 12**: Since $12 < 15$ but $12 > 10$, it becomes the right child of 10.
6. **Insert 17**: Since $17 > 15$ but $17 < 20$, it becomes the left child of 20.
7. **Insert 25**: Since $25 > 15$ and $25 > 20$, it becomes the right child of 20.

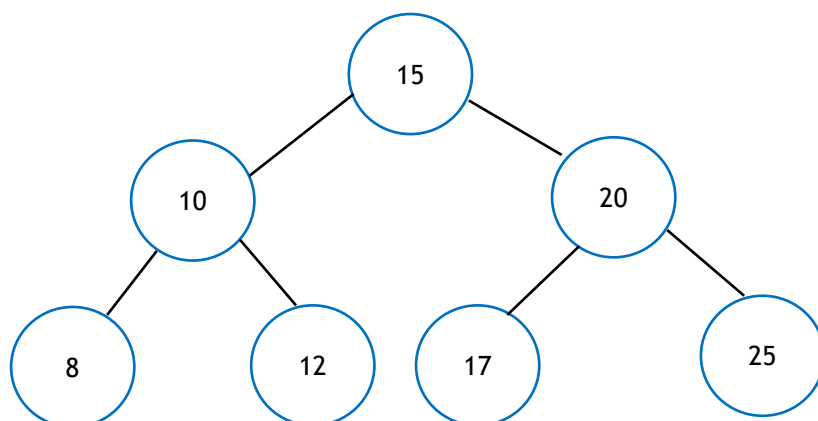


Figure 2: Qualitative Example: Binary Search Tree (BST)

This structure ensures that any search operation can efficiently traverse the tree to locate a specific value.

Tree Traversal Techniques

Traversal is the process of visiting all the nodes in a tree data structure, exactly once, in a systematic way. The primary traversal methods for binary trees are:

1. **Depth-First Traversal (DFT):** Explores as far as possible along each branch before backtracking.
 - **Preorder Traversal (NLR):** Visit the root node first, then recursively traverse the left subtree, followed by the right subtree.
 - **Inorder Traversal (LNR):** Recursively traverse the left subtree first, visit the root node, and then traverse the right subtree. In BSTs, this traversal retrieves values in ascending order.
 - **Postorder Traversal (LRN):** Recursively traverse the left subtree, then the right subtree, and visit the root node last.
2. **Breadth-First Traversal (BFT) or Level Order Traversal:** Visits all nodes at each level before moving to the next level. It starts from the root and visits nodes level by level from left to right (geeksforgeeks, 2025).

Example: Traversing the BST above

- **Preorder Traversal:** 15, 10, 8, 12, 20, 17, 25
- **Inorder Traversal:** 8, 10, 12, 15, 17, 20, 25
- **Postorder Traversal:** 8, 12, 10, 17, 25, 20, 15
- **Level Order Traversal:** 15, 10, 20, 8, 12, 17, 25

Applications of Binary Trees

- **Data Compression:** Binary trees are utilized in algorithms like Huffman coding to compress data efficiently by assigning variable-length codes to input characters based on their frequencies.
- **Expression Parsing:** In compilers, binary trees help in parsing mathematical expressions and programming languages by representing the hierarchical structure of operations.
- **Database Indexing:** Binary trees, particularly BSTs, are used in databases to index data, allowing for quick search, insertion, and deletion operations (Geeksforgeeks, 2024).

Q4: Access whether an Eulerian and Hamiltonian circuit exists in undirected graphs.

Eulerian Circuit

A **Eulerian circuit** is a cycle that traverses every edge of a graph exactly once and returns to the starting vertex. For an undirected graph to have an Eulerian circuit, it must satisfy the following conditions:

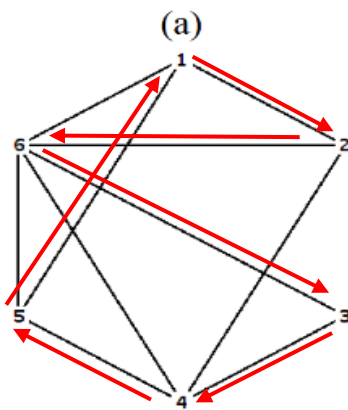
1. **Connectedness:** The graph must be connected, meaning there's a path between any pair of vertices.
2. **Even Degree Vertices:** Every vertex in the graph must have an even degree (an even number of edges incident to it) (geeksforgeeks, 2013).

Hamiltonian Circuit

A **Hamiltonian circuit** is a cycle that visits every vertex of a graph exactly once and returns to the starting vertex. Determining the existence of a Hamiltonian circuit is more complex than for Eulerian circuits, as there isn't a simple, universal criterion applicable to all graphs (GeeksforGeeks, 2012).

(a)

- **Eulerian circuit:** Eulerian circuit for this graph is not possible, because some vertices of the graph have odd degree (1, 2, 5 and 6). (geeksforgeeks, 2013)
- **Hamiltonian circuit:** Hamiltonian circuit for this graph is possible.

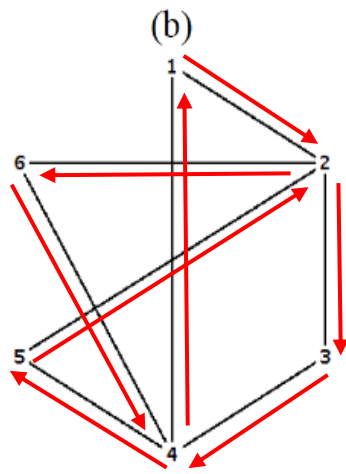


Hamiltonian circuit: 1→2→6→3→4→5→1

Figure 3: (a) Hamiltonian circuit path

(b)

- **Eulerian circuit:** Eulerian circuit for this graph is possible, because all vertices of the graph has even degree (geeksforgeeks, 2013).



Eulerian circuit:
 $1 \rightarrow 2 \rightarrow 6 \rightarrow 4 \rightarrow 5 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 1$

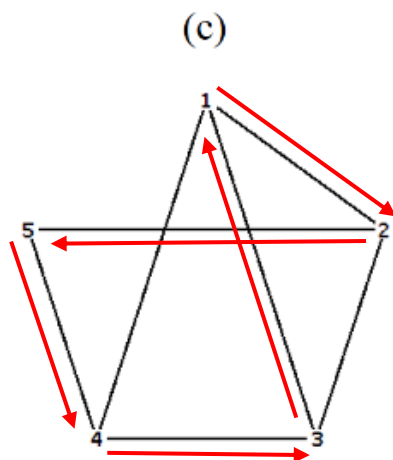
Figure 4: (b) Eulerian circuit path

- **Hamiltonian circuit:** Hamiltonian circuit for this graph is not possible.

(c)

- **Eulerian circuit:** Eulerian circuit for this graph is not possible, some vertices of the graph has odd degree (1, 2, 3 and 4). (geeksforgeeks, 2013)

- **Hamiltonian circuit:** Hamiltonian circuit for this graph is possible.



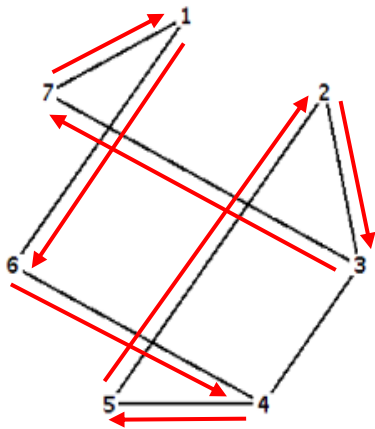
Hamiltonian circuit:
 $1 \rightarrow 2 \rightarrow 5 \rightarrow 4 \rightarrow 3 \rightarrow 1$

Figure 5: (c) Hamiltonian circuit path

(d)

- **Eulerian circuit:** Eulerian circuit for this graph is not possible, some vertices of the graph has odd degree (3 and 4). (geeksforgeeks, 2013)
- **Hamiltonian circuit:** Hamiltonian circuit for this graph is possible.

(d)



Hamiltonian circuit:
 $7 \rightarrow 1 \rightarrow 6 \rightarrow 4 \rightarrow 5 \rightarrow 2 \rightarrow 3 \rightarrow 7$

Figure 6: (d)Hamiltonian circuit path

Q5: Finding the shortest path using Dijkstra's algorithm

Dijkstra's algorithm is a well-known method for finding the shortest paths between nodes in a graph, Developed by Edsger W. Dijkstra in 1956. it operates by assigning a tentative distance value to every node: zero for the initial node and infinity for all other nodes. It then explores the nearest unvisited node, calculates the tentative distances through it to its neighbours, and updates the neighbour's distance if smaller.

This process continues until all nodes have been visited, ensuring the shortest path from the source node to all other nodes is found (wscubetech, 2025).

i) Finding the shortest path and shortest distance from a to z of the following graph.

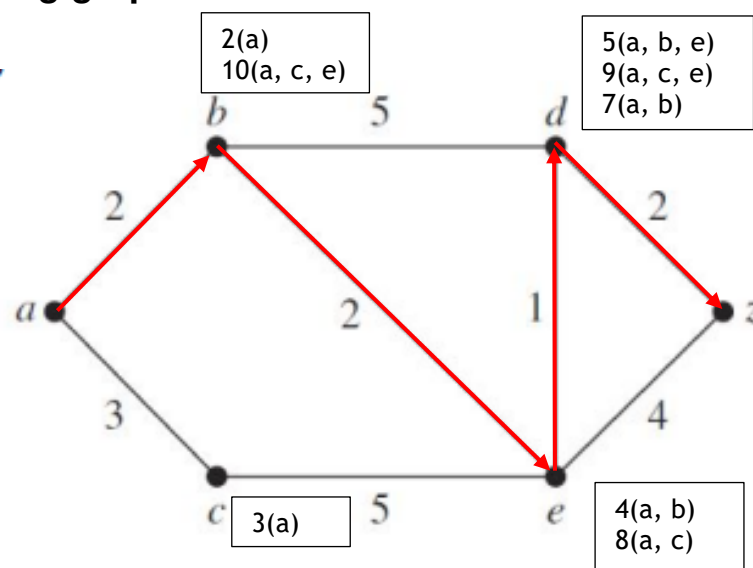


Figure 7: finding the shortest path from a to z

Possible paths from a to z (wscubetech, 2025):

$$a \rightarrow b \rightarrow e \rightarrow z = 8$$

$$a \rightarrow c \rightarrow e \rightarrow z = 12$$

$$a \rightarrow b \rightarrow d \rightarrow z = 9$$

$$a \rightarrow c \rightarrow e \rightarrow d \rightarrow z = 11$$

$$a \rightarrow b \rightarrow e \rightarrow d \rightarrow z = 7$$

The shortest path & shortest distance from a to z is: $a \rightarrow b \rightarrow e \rightarrow d \rightarrow z = 7$

ii) Finding the shortest path and shortest distance from s to t of the following graph.

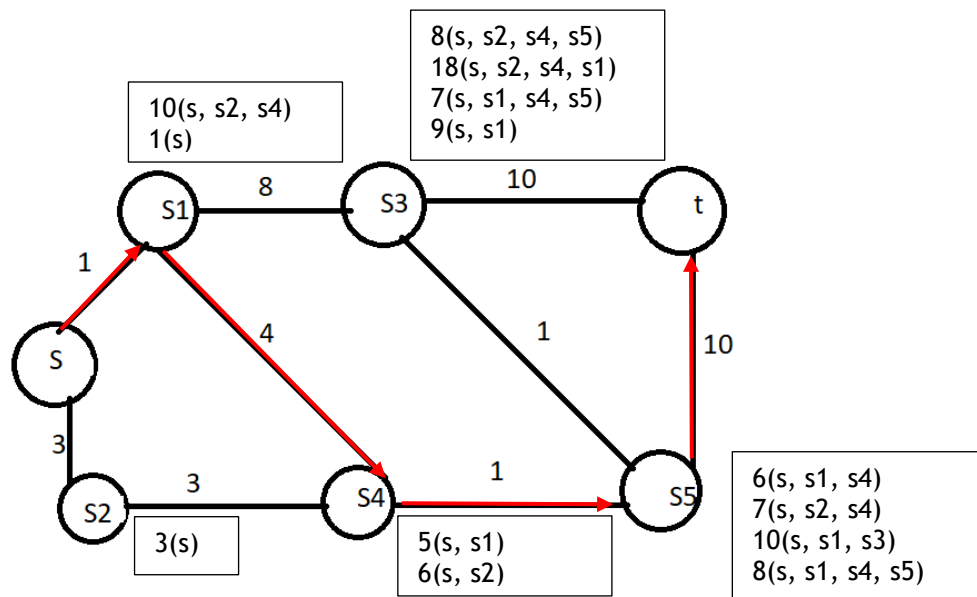


Figure 8: finding the shortest path from s to t

Possible paths from s to t (wscubetech, 2025):

$$s \rightarrow s1 \rightarrow s3 \rightarrow t = 19$$

$$s \rightarrow s1 \rightarrow s4 \rightarrow s5 \rightarrow s3 \rightarrow t = 17$$

$$s \rightarrow s1 \rightarrow s3 \rightarrow s5 \rightarrow t = 20$$

$$s \rightarrow s1 \rightarrow s4 \rightarrow s5 \rightarrow t = 16$$

$$s \rightarrow s2 \rightarrow s4 \rightarrow s5 \rightarrow t = 17$$

$$s \rightarrow s2 \rightarrow s4 \rightarrow s5 \rightarrow s3 \rightarrow t = 18$$

The shortest path & shortest distance from s to t is: $s \rightarrow s1 \rightarrow s4 \rightarrow s5 \rightarrow t = 16$

Q6: Finding the output equation for the logic circuit and prepare the truth table for the output equation.

(Bratty, 2024)

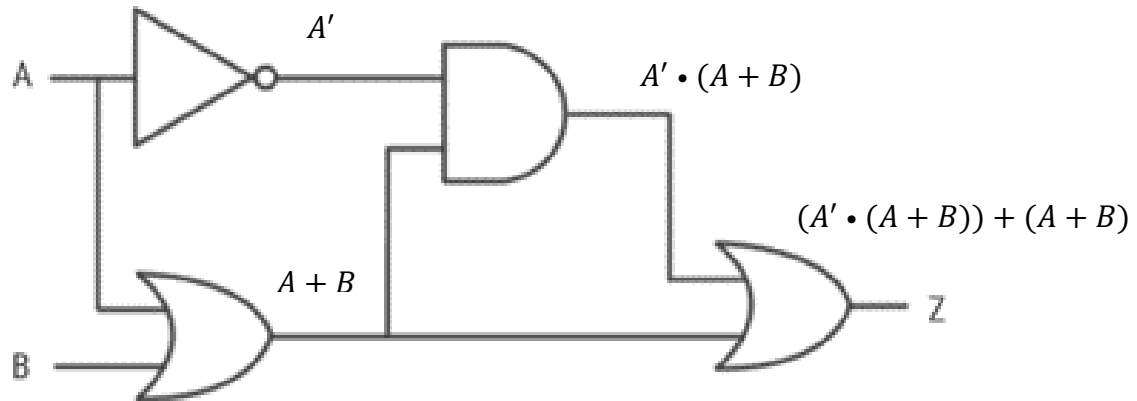


Figure 9: finding the logical circuit equation output of (z)

- Output equation of the logic circuit: $Z = (A' \cdot (A + B)) + (A + B)$
- (Bratty, 2024)

Truth table for the output equation.

A	B	A'	A + B	$A' \cdot (A + B)$	$(A' \cdot (A + B)) + (A + B) = Z$
0	0	1	0	0	0
0	1	1	1	1	1
1	0	0	1	0	1
1	1	0	1	0	1

Table 1: Truth table of Logical circuit (Bratty, 2024)

Q7: Simplifying Boolean expressions.

i) $AB + A(B + C) + B(B + C)$

$AB + AB + AC + BB + BC$ — Distributive Law

$AB + AB + AC + B + BC$ — Idempotent law ($B.B=B$)

$AB + AB + AC + B + BC$ — Idempotent ($AB+AB=AB$)

$AB + AC + B(1 + C)$ — Distributive Law

$AB + AC + B(1)$ — Domination law ($1+C=1$)

$AB + AC + B$ — Identity law ($B.1=B$)

$B(1 + A) + AC$ — Distributive Law

$B(1) + AC$ — Identity Law ($1+A=1$)

$B + AC$ — Identity law ($B.1=B$)

Simplified expression (Storr, 2022) is: $B + AC$

ii) $AB' + A(B + C)' + B(B + C)'$

$AB' + A(B'C') + B(B'C')$ — De Morgan's Law

$AB' + AB'C' + BB'C'$ — Distributive Law

$AB' + AB'C' + 0C'$ — Complement Law ($B.B'=0$)

$AB' + AB'C' + 0$ — Identity Law ($0.C'=0$)

$AB' + AB'C'$ — Identity Law ($AB'C'+0=AB'C'$)

$AB'(1 + C')$ — Distributive Law

$AB'(1)$ — Domination Law ($1+C'=1$)

AB' — Identity Law ($AB'.1=AB'$)

Simplified expression (Storr, 2022) is: AB'

iii) $(AB' (C + BD) + A'B')$

$AB'C + AB'BD + A'B'$	— Distributive Law
$AB'C + A(0)D + A'B'$	— Complement Law ($B'.B=0$)
$AB'C + 0 + A'B'$	— Identity Law ($A.0.D=0$)
$AB'C + A'B'$	— Identity Law ($0+A'B'= A'B'$)
$B'(AC + A')$	— Distributive Law
$B'(A' + C)$	— Absorption Theorem
$A'B' + B'C$	— Distributive Law

Simplified expression (Storr, 2022) is: $A'B' + B'C$

iv) $A'BC + AB'C' + A'B'C' + AB'C + ABC$

$A'BC + ABC + AB'C' + AB'C + A'B'C'$	— Commutative Law
$C(A'B + AB) + AB'(C' + C) + A'B'C'$	— Distributive Law
$C(A'B + AB) + AB'(1) + A'B'C'$	— Complement Law ($C'+C=1$)
$C(A'B + AB) + AB' + A'B'C'$	— Identity Law ($AB'.1=AB'$)
$CB(A' + A) + AB' + A'B'C'$	— Distributive Law
$CB(1) + AB' + A'B'C'$	— Complement Law ($A'+A=1$)
$CB + AB' + A'B'C'$	— Identity Law ($CB.1=CB$)
$CB + B'(A + A'C')$	— Distributive Law
$CB + B'(A + C')$	— Absorption Theorem
$CB + AB' + B'C'$	— Distributive Law

Simplified expression (Storr, 2022) is: $CB + AB' + B'C'$

Q8: Truth Table Analysis and SOP Derivation for an Electronic Door System.

Given Conditions:

1. The door opens if **Sensor A is activated** ($A = 1$).
2. The door opens if **both sensors A and B are activated** ($A = 1$ and $B = 1$).

Since condition 2 already includes condition 1, we can simplify this to:

- The door opens **whenever A is 1** (regardless of B's value).

8.1 Truth Table:

A	B	Door (Output)
0	0	0
0	1	0
1	0	1
1	1	1

Table 2: Truth table Electronic Door System (GeeksforGeeks, 2023).

The derived output equation from the truth table: $A\bar{B} + AB$

8.2 Finding the final equation using Sum of Products (SOP)

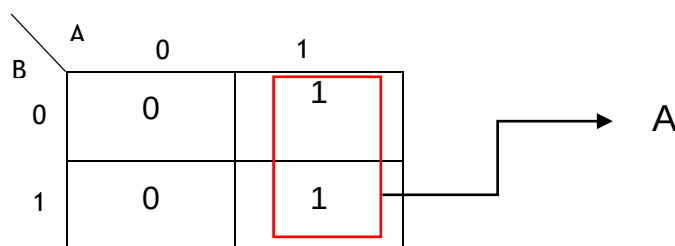


Table 3: Karnaugh Map (SOP) (GeeksforGeeks, 2015)

8.3 logic gate based digital circuit for the final output equation.



Q9: real-world binary problem in the application of Boolean algebra

Boolean algebra is fundamental in digital circuit design, enabling the representation and simplification of logical expressions using binary variables and operations such as AND, OR, and NOT. A practical application of Boolean algebra is in the control logic of air conditioning systems, where decisions to turn the system on or off depend on specific conditions (GeeksforGeeks, 2021).

9.1 Example Scenario

Air Conditioner Control Scenario:

Consider an air conditioning system that operates based on two conditions:

1. **Temperature Status (T):**
 - $T = 1$: Temperature is above the set limit.
 - $T = 0$: Temperature is below the set limit.
2. **Occupancy Status (P):**
 - $P = 1$: Person is present in the room.
 - $P = 0$: Room is unoccupied.

The air conditioner (AC) should function as follows:

- **Turn ON (AC = 1):** When the temperature is above the limit ($T = 1$) *and* a person is present ($P = 1$).
- **Turn OFF (AC = 0):** In all other scenarios.

9.2 Truth Table

Temperature (T)	Person Present (P)	Air Conditioner (AC)
0	0	0
0	1	0
1	0	0
1	1	1

Table 4: Truth table for the Air Conditioner Control Scenario (Storr, 2013)

9.3 Boolean Expression

From the truth table, the Boolean expression for the AC control is:

$$AC = T \cdot P$$

This expression indicates that the AC will turn on ($AC = 1$) only when both T and P are 1 (GeeksforGeeks, 2021).

9.4 Logic Circuit Diagram

The corresponding logic circuit consists of an AND gate with inputs T and P, and the output representing the AC control signal.

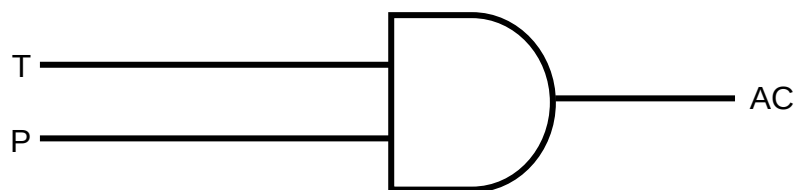


Figure 10: Logic circuit for Air Conditioner Control Scenario

Inputs:

- T: Indicates if the temperature is above the limit.
- P: Indicates if a person is present in the room.

AND Gate: Outputs 1 (turns AC on) only when both inputs are 1, aligning with the control logic requirements (Avy, 2016).

9.5 Real-World Implementation

In practical applications, sensors detect room temperature and occupancy:

- **Temperature Sensor:** Monitors ambient temperature and sets T accordingly.
- **Occupancy Sensor:** Detects presence in the room and sets P accordingly.

These sensors interface with a control unit that processes the inputs using the defined logic to control the AC system.

This example illustrates how Boolean algebra facilitates the design of efficient and effective control systems in real-world applications (Binary Problem), such as managing air conditioning based on environmental and occupancy conditions.

Q10: Colorize following provincial map of Sri Lanka using five-colour theorem.

This theorem states that any planar graph (a graph that can be drawn on a plane without edges crossing) can be coloured using no more than five colours, ensuring that no two adjacent regions share the same colour. This concept applies to maps where regions (provinces) sharing a border must have different colours (GeeksforGeeks, 2023).

To colorize a provincial map of Sri Lanka using the five-colour theorem, we have to follow these steps:

- 1. Identify Adjacent Provinces:** Sri Lanka consists of nine provinces. To apply the five-colour theorem effectively, determine which provinces share borders. This adjacency information is crucial for assigning colours.
- 2. Assign Colours Using the Five-Colour Theorem:**
 - **Start with a Province:** Choose any province and assign it the first colour.
 - **Proceed to Adjacent Provinces:** Assign colours to neighboring provinces, ensuring no two adjacent provinces share the same colour. Utilize up to five distinct colours as needed.
 - **Continue the Process:** Move systematically through the map, applying colours based on adjacency and the five-colour limit.

By following these steps, we can effectively colorize the provincial map of Sri Lanka in accordance with the five-colour theorem, ensuring that no two adjacent provinces share the same colour.

Here is the colorized provincial map of Sri Lanka using five-colour theorem.

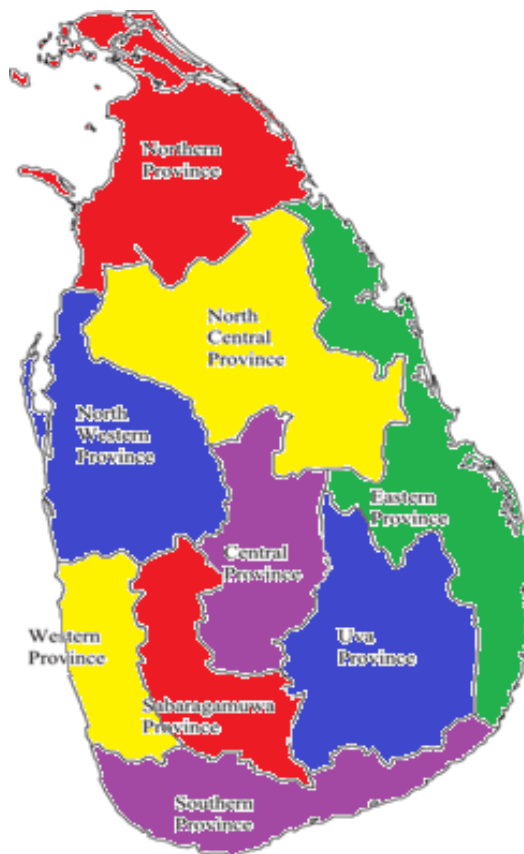


Figure 11: colorized provincial map of Sri Lanka using five-colour theorem

Q11: analyze the properties of addition, subtraction, multiplication, and division concerning their associative and commutative nature.

In the set of natural numbers ($\mathbb{N} = \{1, 2, 3, \dots\}$), we can analyze the properties of addition, subtraction, multiplication, and division concerning their associative and commutative nature.

(a) Associative Binary Operation

An operation $*$ is **associative** if, for all a , b , and c in the set, the equation $(a * b) * c = a * (b * c)$ holds.

- **Addition (+):** Addition is associative in $\mathbb{N}$. For any natural numbers a , b , and c : $(a + b) + c = a + (b + c)$
✓ *Example:* $(2 + 3) + 4 = 5 + 4 = 9$ and $2 + (3 + 4) = 2 + 7 = 9$
(BrainRouter, 2020).
- **Subtraction (−):** Subtraction is not associative in $\mathbb{N}$. For example: $(5 - 3) - 1 = 2 - 1 = 1$, whereas $5 - (3 - 1) = 5 - 2 = 3$. Since $1 \neq 3$, subtraction fails the associative property (BrainRouter, 2020).
- **Multiplication (×):** Multiplication is associative in $\mathbb{N}$. For any natural numbers a , b , and c : $(a \times b) \times c = a \times (b \times c)$
✓ *Example:* $(2 \times 3) \times 4 = 6 \times 4 = 24$ and $2 \times (3 \times 4) = 2 \times 12 = 24$
(BrainRouter, 2020).
- **Division (÷):** Division is not associative in $\mathbb{N}$. For example: $(8 \div 4) \div 2 = 2 \div 2 = 1$, whereas $8 \div (4 \div 2) = 8 \div 2 = 4$. Since $1 \neq 4$, division fails the associative property (BrainRouter, 2020).

(b) Commutative Binary Operation

An operation $*$ is **commutative** if, for all a and b in the set, the equation $a * b = b * a$ holds.

- **Addition (+):** Addition is commutative in $\mathbb{N}$. For any natural numbers a and b : $a + b = b + a$
✓ *Example:* $2 + 3 = 5$ and $3 + 2 = 5$ (BrainRouter, 2020).
- **Subtraction (-):** Subtraction is not commutative in $\mathbb{N}$. For example: $5 - 3 = 2$, whereas $3 - 5 = -2$. Since $2 \neq -2$, subtraction fails the commutative property (BrainRouter, 2020).
- **Multiplication (×):** Multiplication is commutative in $\mathbb{N}$. For any natural numbers a and b : $a \times b = b \times a$
✓ *Example:* $2 \times 3 = 6$ and $3 \times 2 = 6$ (BrainRouter, 2020).
- **Division (÷):** Division is not commutative in $\mathbb{N}$. For example: $8 \div 4 = 2$, whereas $4 \div 8 = 0.5$. Since $2 \neq 0.5$, division fails the commutative property (BrainRouter, 2020).

Operation	Associative	Commutative
Addition (+)	Yes	Yes
Subtraction (-)	No	No
Multiplication (×)	Yes	Yes
Division (÷)	No	No

Table 5: summary of Properties of Binary Arithmetic Operations (BrainRouter, 2020).

Q12:

(a) Determine the order of the group and the orders of its subgroups for $G = \mathbb{Z}_6$ (the integers modulo 6 under addition).

1. Order of the Group $\mathbb{Z}_6$:

The group $\mathbb{Z}_6$ consists of integers modulo 6 under addition, meaning its elements are $\{0, 1, 2, 3, 4, 5\}$. The **order** of a group is defined as the number of elements in the set. Therefore, the order of $\mathbb{Z}_6$ is 6 (Fraleigh, 2025).

2. Orders of Subgroups in $\mathbb{Z}_6$:

According to Lagrange's Theorem, the order of any subgroup must divide the order of the entire group. Given that the order of $\mathbb{Z}_6$ is 6, the possible orders for its subgroups are the divisors of 6: 1, 2, 3, and 6.

- **Subgroup of Order 1:** The trivial subgroup $\{0\}$, containing only the identity element.
- **Subgroup of Order 2:** $\{0, 3\}$. Adding 3 to itself yields 0 modulo 6, demonstrating closure under addition.
- **Subgroup of Order 3:** $\{0, 2, 4\}$. Adding 2 repeatedly cycles through these elements: $2 + 2 = 4$, and $4 + 2 = 6 \equiv 0 \pmod{6}$.
- **Subgroup of Order 6:** The group itself, $\mathbb{Z}_6$, which includes all elements $\{0, 1, 2, 3, 4, 5\}$.

By addressing each criterion, we've determined the orders of the group and its subgroups in $\mathbb{Z}_6$ (Macauley, 2014).

(b) Validate that the set of non-zero real numbers with the operation of multiplication is a commutative group.

To validate that the set of non-zero real numbers ($\mathbb{R} \setminus \{0\}$) under multiplication forms a commutative group, we need to verify the group axioms: closure, associativity, identity element, inverses, and commutativity (proofwiki, 2022).

- **Closure:** For any $a, b \in \mathbb{R}^*$, the product $a \times b$ is also a non-zero real number. Therefore, $\mathbb{R}^*$ is closed under multiplication.
- **Associativity:** Multiplication of real numbers is associative; that is, for any $a, b, c \in \mathbb{R}^*$, $(a \times b) \times c = a \times (b \times c)$.
- **Identity Element:** The number 1 serves as the multiplicative identity in $\mathbb{R}^*$, since for any $a \in \mathbb{R}^*$, $a \times 1 = a = 1 \times a$. (proofwiki, 2021)
- **Inverse Element:** Every non-zero real number a has a multiplicative inverse $a^{-1} = \frac{1}{a}$ such that $a \times a^{-1} = 1$.
- **Commutativity:** Multiplication of real numbers is commutative; for any $a, b \in \mathbb{R}^*$, $a \times b = b \times a$.

Since all group axioms are satisfied, $\mathbb{R}^*$ under multiplication is indeed a commutative group (proofwiki, 2021).

By addressing each criterion, we've validated that the set of non-zero real numbers under multiplication forms a commutative group.

Conclusion

This assignment has provided a comprehensive overview of how various discrete mathematics concepts can be applied to practical computing problems. From calculating cardinalities and inverting functions to analyzing graphs for shortest paths and simplifying Boolean expressions, each topic reinforces critical skills necessary for system design, algorithm development, and software testing. Furthermore, the discussion on abstract algebra highlights the underlying structure and symmetry within computing logic. Overall, this exploration has deepened my appreciation for the importance of discrete mathematics in the field of computing and enhanced my ability to model and solve complex problems logically and efficiently.

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